MIL-A-21059A(Aer)
29 September 1959
Superseding
MIL-A-21059(Aer)
15 August 1957

MILITARY SPECIFICATION

AEROLOGICAL DISPLAY EQUIPMENT

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification covers the following items, of one grade, and of type and size as shown on applicable drawings:

(a) Display Board, Meteorological Data ML-529( )/U
(b) Display Board, Meteorological Data ML-530( )/U
(c) Table, Observer, Meteorological Data ML-531( )/U
(d) Table, Forecaster, Meteorological Data ML-532( )/U
(e) Plotting Table, Meteorological Data ML-533( )/U
(f) Cabinet, Filing, Meteorological Data CY-2127( )/UM
(g) Cabinet, Filing, Meteorological Data CY-2128( )/UM
(h) Cabinet-Display, Meteorological Data CY-2129( )/UM
(i) Cabinet-Display, Meteorological Data CY-2130( )/UM

1.2 Nomenclature. The parentheses appearing in the nomenclature will be dropped or replaced by a letter, to indicate the particular design, upon application to the procuring activity.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on the date of invitation for bids, forms a part of this specification.

Specifications.

Federal.

GG-S-236	Sharpeners, Pencil
HH-C-576	Cork-Composition; Gasket and Sheet
QQ-A-318i	Aluminum-Alloy (52S)(Aluminum-Magnesium-Chromium); Plate and Sheet
QQ-A-325;	Aluminum-Alloy (61S)(Aluminum-Magnesium-Silicon-Chromium); Bars, Rods, Shapers and Wire
QQ-A-270	Aluminum-Alloy (61S) Bars, Rods, and Shapers, extruded
QQ-S-633	Steel; Carbon (Low Carbon, Sheets and Strips)
QQ-S-766	Steel; Corrosion-resisting; Plates, Sheets, Strips, and Structural Shapes
TT-P-636	Primer, Syntnetic for Ferrous Metals
UU-P-271	Paper, wrapping, waterproofed kraft
PPP-B-601.	Boxes, wood, cleated-plywood

Military.

MIL-P-116	Preservation, methods of
MIL-E-5556	Enamel, Synthetic, Semi-glass #621

FSC 6660

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MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base)
MIL-N-18307	Nomenclature and nameplates for airborne electronic and associated equipment
MIL-P-21105	Plastic, sheet acrylic, heat resistant (utility grade)

Standards:

ML-STD-129B	Marking for shipment on storage

Drawings:

Bureau of Aeronautics

Buaer No.	Applicable Nomenclature	Discontinued Nomenclature	Page No.
56A13J1	1. Table, Forecaster, Meteorological Data ML-532( )/U	Forecaster's Chart & Draft Board, Type WX1-00	8
56A13J5	2. Plotting Table, Meteorological Data ML-533( )/U	Spotter's Chart & Draft Board, Type WX2-00	9
56A13J6 Rev. 2	3. Table, Observer, Meteorological Data ML-531( )/U	Observer's Technical Stand, Type WX3-00	10
56A13J8 Rev. 1	4. Display Board, Meteorological Data ML-530( )/U	Flight Display Board, Type WX4A-00	11
56A13J9	5. Display Board, Meteorological Data ML-529( )/U	Flight Display Board, Type WX4B-00	12
56A13J10	6. Cabinet-Display, Meteorological Data CY2129( )/UM	Weather Sequence Panel, Type WX5A-00	13
56A13J13	7. Cabinet-Display, Meteorological Data CY2130( )/UM	Weather Sequence Panel, Type WX5B-00	14
56A13J15	8. Cabinet, filing, Meteorological Data CY2127( )/UM	Synoptic Map Filing Compartment, Type WX6A-00	15
56A13J17	9. Cabinet, filing, Meteorological Data CY2128( )/UM	Synoptic Map Filing Compartment, Type WX6B-00	16

Bureau of Aeronautics Drawings: The drawings furnished hereunder as part of this specification - for general guidance in design only. Construction details and method of fabrication are the responsibility of the contractor. All question of interpretation and clarification of minor construction and assembly details incident to the production of the equipment are to be resolved locally by the cognizant naval inspector subject to determination by the inspector that the resulting items are adequate to insure serviceable, well constructed equipment. Any change in basic design shall be referred to the Chief of the Bureau of Aeronautics (AY-2).

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.)

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3. REQUIREMENTS

3.1 Materials. The materials shall conform in all respects to the specifications and drawings listed in Section 2 and as specified hereinafter.

3.1.1 Sheet, aluminum. Shall conform to the requirements of Specification QQ-A-318b(1), 1/4 hard condition (52S-1/4H). The finish of the flat sheet where painted may be a mill-finish, whereas the parts to be exposed and finished naturally shall be a bright finish with a buffed polish, as called for on applicable drawings.

3.1.2 Rivets, aluminum. Shall conform to the requirements of Specification QQ-A-325a(2), heat-treated condition (61S)

3.1.3 Pipe, aluminum. Shall conform to the chemical and physical requirements of Specification QQ-A-270(1) except that it shall be classed as extra-heavy pipe (61S), sizes and wall thickness as shown on drawings.

3.1.4 Steel, stainless. For the Guard Rail and Binding, shall conform to the requirements of Specification QQ-S-766a(1), Class I, Finish 2, commercial polish, Tampico brushed finish.

3.1.5 Steel, plain. For construction of the suspension slide, hardware, and miscellaneous plates and reinforcement bars shall conform to the requirementsof Specification QQ-S-636(2) and shall be zinc-coated with a coating not less than 0.0003 in. thickness. Carbon content shall be not less than .08%.

3.1.6 Tubing, steel. The steel shall be 1.000 in. outside diameter by 0.065 in. plus or minus 0.004 in. wall thickness and shall be either seamless drawn, or welded and drawn, stainless steel tubing containing not less than 18 percent chromium and 8 percent nickel; or shall be seamless drawn, or welded and drawn, cold-rolled steel tubing with a chromium plated finish consisting of plated coatings of the following minimum thickness:

Copper	-	0.0005 in.
Nickel	-	0.001 in.
Chromium	-	0.00002 in.

3.1.7 Steel-spring. Both wire and strip for the spring clips shall be of standard commercial quality suitable for the purpose intended and shall not be subject to brittleness and cracking when flexed in the natural use to which these parts will be put. The finish on the springs shall be a baked black enamel finish.

3.1.8 Cork, sheet. Shall conform to the requirements of Class 3 quality, of Specification HH-C-576(1) size of sheet and thickness shall be as indicated on drawing.

3.1.9 Bolts and nuts. Where not exposed, shall be steel, zinc-coated, and where exposed shall be brass or bronze, with a satin chrome or nickel finish. All bolts and nuts shall be free fit, cut threads, and shall conform to standard commercial practice.

3.1.10 Screws, machine. Where not exposed shall be steel zinc-coated, and where exposed shall be of brass or bronze, satin chrome or nickel finish. Screws, machine, shall be hexagonal head, fine thread, and shall conform to standard commercial practice.

3.1.11 Plastic, acrylic sheet. Shall conform to the requirements of Specification MIL-P-21105A. The material shall be frosted on the underside where indicated on applicable drawings.

3.1.12 Fixtures, fluorescent lighting. The lighting fixtures shall conform to the design as indicated on the drawings and shall be High Grade, standard commercial quality, suitable for the purpose intended. Likewise the switches, wire and light bulbs shall be of appropriate grade, design and quality for the purpose intended.

3.1.13 Casters and glides. Shall be standard commercial items conforming to the size and design shown on applicable drawings and of a quality suitable for the purpose intended. All glides and casters shall be installed in the leg assemblies by the manufacturer prior to shipment.

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..14 Hardware. Handles, knobs, hinges, etc., shall be constructed of materials as indicated the drawings and of similar design and shall be of a quality suitable for the purpose intended. ere aluminum is used, it shall be finished with a soft satin finish, and where brass or bronze used, the finish shall be satin chrome or nickel.

1.15 Pencil Sharpeners. Shall conform to requirements of Specification CC-S-236b(1) Type II.

1.16 Clips, spring. Shall be of standard commercial design and of suitable quality for the pose intended.

2 Design and Construction. Design and construction shall conform to the referenced drawings paragraph 2 and detailed requirements stated below.

2.1 Fabrication. All fixtures shall be of knock-down construction or assemblies, and shall be pable of being set up readily with simple tools. All clearances shall be close in order to make able and rigid fixtures when set up for use. All fixtures shall be machined accurately and free om burrs and rough edges which will hamper assembly as illustrated on the assembly charts (see ragraph 3.4). The application of hardware and other component parts shall be accurately accomplished

2.2 Welding. The welding may be done by either a gas or electric-arc process or a combination the two methods, however, all welding shall be in accordance with established welding procedures d by welders trained in the technique of welding aluminum. The parts to be welded shall be operly prepared, to insure sound welds of maximum strength, and perfectly aligned into correct sitions. Welding shall be done with welding rods or electrodes containing not less than 95 rcent aluminum, and not more than 5 percent silicon. All welds shall show good fusion and soundness, ee from cracks, porosity, gas pockets or under-cutting. Unexposed welds that do not require inding shall have good surface appearance with fair uniformity and be free from surface holes and atter. Where spot welding is used, the spots shall be not over three inches on center (see ragraph 4.5.1 for Tests). All welds exposed to view shall be ground smooth. As soon as acticable after welding, all welds shall be thoroughly cleaned to remove the residual deposits the aluminum welding flux. Welded parts shall be immersed in a 5 percent sulphuric acid bath ld at 140°F to 160°F for 5 to 10 minutes, removed and immersed in and thoroughly rinsed with hot cold water, in order to prevent corrosion on metal surfaces.

3. Finish.

3.1 Cleaning and preparation of metal. All surfaces shll be thoroughly cleaned to remove all aces of dirt, oil, grease, grit, rust or foreign matter.

(a) Aluminum. Shall be prepared for painting by applying a phosphoric acid treatment in cordance with Specification MIL-M-10578a.

(b) Steel, plain. All steel parts not chrome or nickel finish, shall be given a zinc- ating as specified in paragraph 3.1.5.

3.2 Primer.

(a) Aluminum. After preparation of aluminum as outlined above, all surfaces shall be iven a coat of primer conforming to the requirements of Specification MIL-P-8585. The dry ilm thickness for primer shall be not less than 0.4 mils or more than 0.6 mils at any point. olor Y - Yellow primer shall be used.

(b) Steel, plain. After preparation of steel as outlined above, all surfaces shall be iven a coat of primer conforming to the requirements of TT-P-636b.

3.3 Enamel. A double coat or two single coats of semi-gloss enamel (color to match the sample anel furnished by the contracting office) conforming to Specification MIL-E-5556A, shall then be pplied and baked at a temperature of 300°F for not less than 30 minutes, or an equivalent baking. eriod at a higher or lower temperature. The final, dry film thickness for the enamel shall be ot less than 1.2 mils.

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3.4 Assembly charts - illustrations. An exploded illustration of each fixture shall be made for the purpose of detail explanation of the assembly of the fixture in the field. Charts or diagrams shall be plainly drawn and marked in order to make the assembly with the least possible confusion. The sheet shall be of suitable size to clearly illustrate this information and shall be on gummed paper or glued to an unexposed flat section of the applicable fixture (such as the drawer bottom or inside door panel).

3.5 Plastic acrylic sheet - information. The following information shall be printed on a label and adhered by gummed paper tape to both sides of the plastic sheet:

Cleaning and Polishing

Cleaning. To clean the plastic, wash with plenty of soap and water using the bare hand to feel and dislodge any foreign matter. A soft cloth, sponge or chamois may be used but only as a means of carrying water to the plastic. Dry with a clean damp chamois. Rubbing with a dry cloth builds up an electostatic charge on the plastic so that it attracts dust particles from the air; wiping with a damp chamois will remove this charge as well as the dust and is, therefore, recommended. Remove oil and grease by rubbing lightly with a cloth wet with kerosene or hexane.

Do not use gasoline, alcohol, benzene, acetone, carbon tetrachloride, fire extinguisher or de-icing fluid, lacquer thinners or glass window cleaning sprays, as they will soften the plastic and may cause crazing.

If, after removing dirt and grease, no great amount of scratching is visible, the plastic should be waxed with a good grade of commercial wax. These waxes will fill in minor scratches and help prevent further scratching. The wax should be applied in a thin, even coat and brought up to a high polish by rubbing lightly with a dry, soft-cloth such as Canton or Domet flannel.

Polishing. Scratches too deep to be removed by the hand application of cleaners are often readily removed by buffing.

Best buffing results are obtained with very soft open type of cotton buffing wheels made of Canton or Domet flannel, cotton or muslin. There are a number of standard commercial buffing compounds satisfactory for use with plastic usually composed of very fine alumina or similar "abrasive" in combination with wax, or grease binders and polishing tallow. Both are available in the form of bars or tubes for convenience in applying to the buffing wheel. These compounds are sometimes referred to as "coloring" compounds and the finest grade is as coarse as required for plastic.

If the buffing wheels have been used before, remove any hardened tallow by running them against a sharp metal edge. Start the buffing wheel spinning and apply the stick of tallow to the wheels for a few seconds. Then apply the edge of spinning wheel to the plastic surface very lightly. Keep it moving over the surface and put only light pressure on the buff. Excessive pressure may heat and soften the plastic. Buff along and across any scratch and continue buffing until scratches have been removed. Finally, wipe the buffing compound from the plastic and apply a coat of wax.

If scratches are too deep to be removed by buffing alone, it is sometimes necessary to use sandpaper. However, sanding should not be employed when ordinary buffing can be used to remove the blemish. Further, it should not be used unless some type of mechanical buffing equipment is available, since hand polishing is not sufficiently effective to restore polish to a sanded surface.

3.6 Nameplate and Nomenclature. The nomenclature and type numbers stated in paragraph 1.1 are for preliminary identification only and shall not be used on nameplates. Nomenclature shall be obtained as soon as possible after receipt of contract in accordance with the applicable provisions of Specification MIL-N-18307(ASG). Procedures stated for production contracts shall be followed. All general provisions and requirements indicated for non-airborne equipment shall apply. All requests or submissions of materials shall be directed to the Aerology Division (AY-2) of the Bureau of Aeronautics in lieu of Aer-EL-419 wherever indicated in the specification. Each item of equipment listed in paragraph 1.1 shall be considered a major unit of equipment for nomenclature and nameplate purposes. No overall set nomenclature or nameplates are required and major unit nameplates shall omit the line referring to an overall set.

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3.7 Workmanship. Fixtures, weather station, shall be fabricated by those experienced in the production of sheet metal and metal fabricated products, including forming, riveting, welding and finishing. Each fixture shall be well-made and free from any defects which may affect appearance or serviceability. Rivets shall be properly and tightly peened, and spot welds clean and soundly made.

4. QUALITY ASSURANCE PROVISION

4.1 Samples for approval. The initial samples of production, in all respects, representative of items to be furnished, shall be subject to inspection and approval by representatives of the contracting officer. Contractor shall advise Chief, Bureau of Aeronautics (AY-2) via cognizant Inspector two weeks prior to scheduled availability of equipments for such inspection. When the initial production samples are approved, they shall be retained by the contractor throughout production of the contract as the inspector's sample. These samples may be included in the quantity required for the last shipment.

4.2 Sampling. Samples of any material, components, etc., not furnished by the United States Government, entering into the manufacture of the articles covered herein, may be selected from time to time by the Government inspector and carefully examined and tests made to determine if they are in accordance with this specification.

4.3 Inspection. Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery designated in the contract or purchase order of the procuring agency. In case of factory inspection, every facility shall be afforded inspectors by the manufacturer for the prosecution of their work.

4.4 Contractor's inspection. The contractor shall inspect each fixture for compliance with this specification prior to submitting them to the United States Government for final inspection.

4.5 Tests.

4.5.1 Welds. Test samples shall be prepared by the contractor from time to time as required by the inspector. All welds shall be made under production methods, and shall conform with the requirements as stated in paragraph 3.2.2. Spot welds shall be periodically selected and submitted to the following test: Samples shall be separated to assure soundness of the welds by pulling "buttons" of either welded material.

5. PREPARATION FOR DELIVERY

5.1 Overseas Packing.

5.1.1 Each fixture (comprising one complete unit) shall be disassembled for overseas shipment and packed in a specially designed wood cleated box, constructed to fit the component parts in a compact manner. Boxes shall be constructed in accordance with the weight of contents as foll

5.1.2 Fixtures weighing from 60 lbs. up to and including 175 lbs. shall be packed in Style "A" boxes of 2-ply veneer craft board reinforced with intermediate cleats in accordance with all the requirements of Specification PPP-B-601, Type 1 or 2.

5.1.3 Fixtures weighing over 175 lbs. shall be packed in Style "A" boxes of 3-ply veneer craft board reinforced with intermediate cleats in accordance with requirements outlined in Specifica PP-B-601, Type 1 or 2.

5.1.4 Component parts of each fixture (assemblies or units), similar in size and shape, shall nested or grouped, wrapped, packaged, and securely fastened together to form units for compact packing. Cushioning material such as excelsior pads, single faced corrugated, cellulose waddi or other suitable material shall be provided to prevent projecting edges from marring other component parts or rupturing the waterproof barriers.

5.1.5 The following component parts shall be protected as specified:

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Plexi-Glass

Plexi-Glass panels, installed in the appropriate fixture as indicated by the drawing, shall be fully protected with sheets of double faced corrugated fiberboard and then be wrapped with waterproof barriers material, Specification UU-P-271.

Masonite Tops

Masonite Tops shall be protected and wrapped in the same manner as specified for Plexi-Glass panels.

Light Bulbs and Reflectors

Light bulbs of each fixture shall be packed in single faced corrugated sleeves or tubes and the required amount of bulbs for each reflector shall be packed inside the reflector. Firmly packed excelsior or cellulose wadding shall cushion the light bulbs within the reflector in such manner to prevent movement and damage in transit. Each lighting fixture shall be preserved and packaged in accordance with method 1A14 of Specification MIL-P-116.

Small Component Parts

Small component parts such as screws, bolts, nuts and adjustable gliders shall be packed in a folding, set-up or corrugated fiberboard box and then wrapped with waterproof barrier material conforming to Specification UU-P-271.

5.1.6 The assemblies or component parts of each fixture shall be arranged in the shipping container with the larger and heavier pieces to the bottom of the box and the smaller and lighter and more fragile pieces to the top. Wood bracing and blocking shall be provided to secure and immobilize the heavier parts and shall be secured to the wood battens and not to panels, with cement-coated nails. Suitable water resistant cushioning material shall be tightly wadded to fill all void spaces in the container.

5.1.7 All waterproof barrier material specified for protective wrapping shall conform to Specification UU-P-271. All seams, joints and folds shall be secured, sealed with a water-resistant adhesive to provide protection equal to that afforded by the barrier material.

5.1.8 The shipping boxes shall be reinforced with steel straps to conform with the requirements outlined in the Appendix of Specification PP-B-601.

5.2 Domestic Packing. Aerological Display Equipment shall be wrapped, packaged, and packed in the same manner as outlined in the overseas packing except that the waterproof qualities of the barrier material, adhesive, and cushioning materials specified for portective wrapping, etc., will not be required.

5.3 Marking. Shipping containers shall be marked in accordance with the requirements outlined in Standard MIL-STD-129.

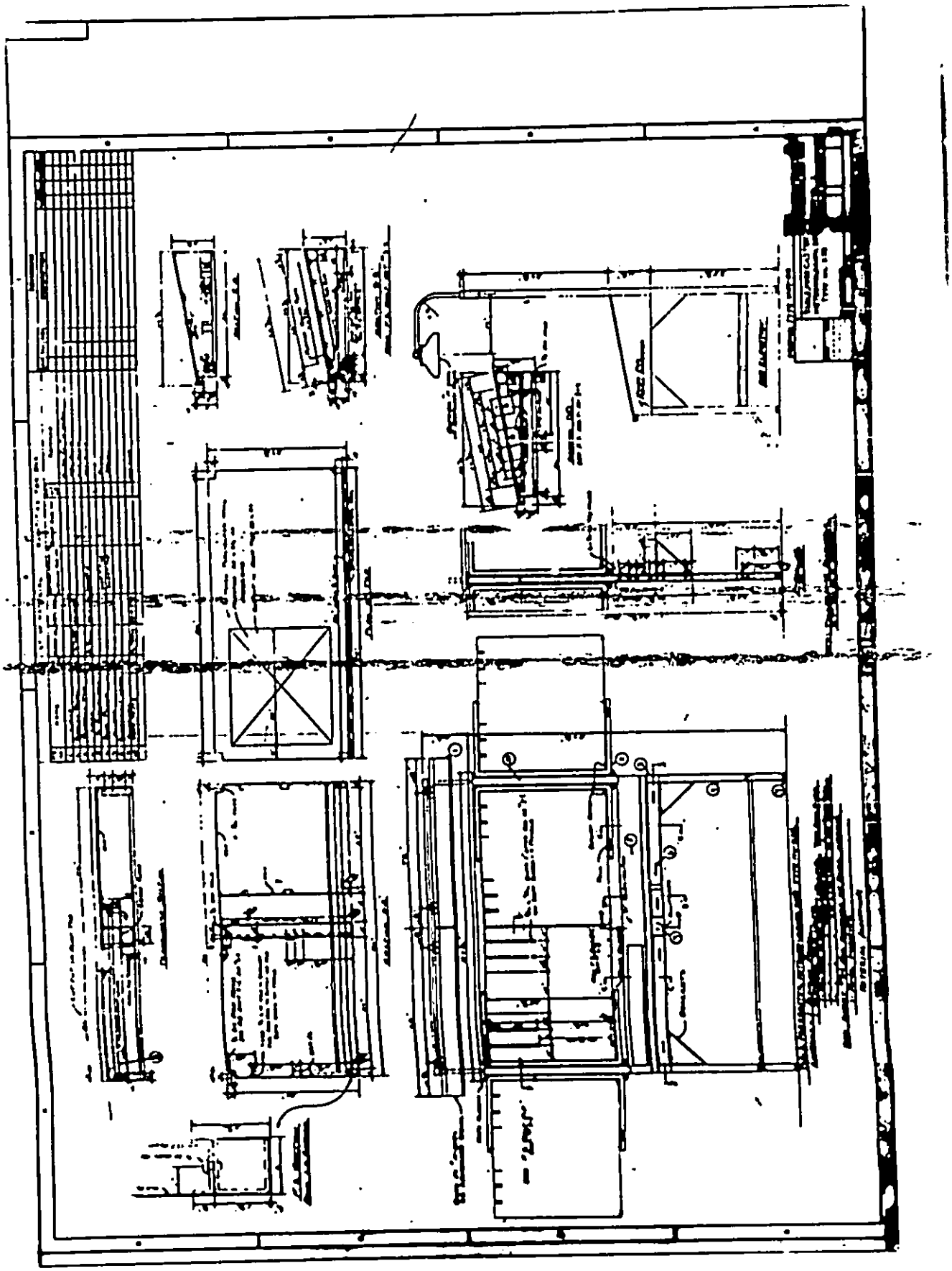
5.3.1 Special Marking. Arrows shall be prominently marked on all sides of the box to indicate the position in which the box is is to be handled in transit. The word "UP" shall appear at the point of each arrow in letters at least 2 in. high.

6. NOTES

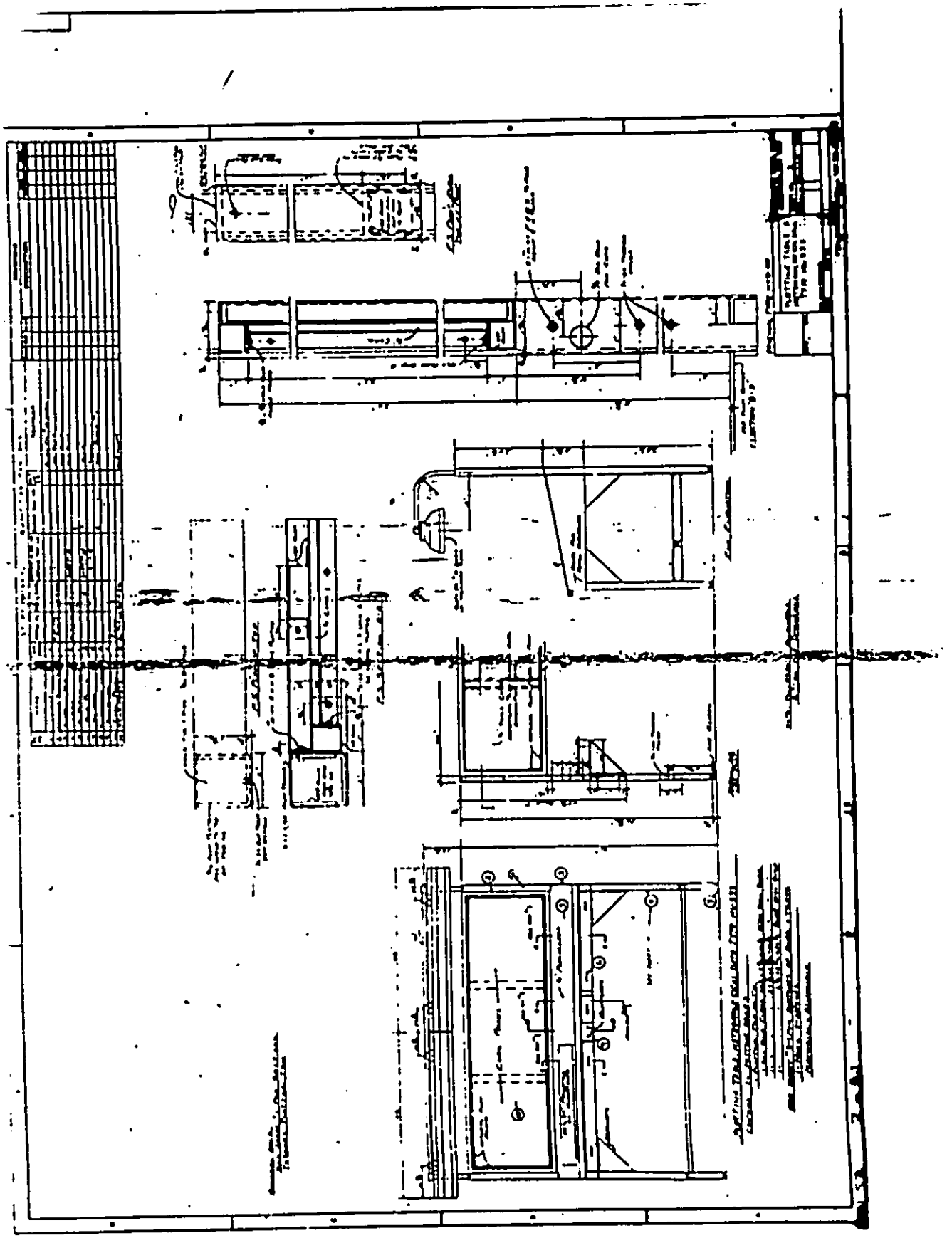
6.1 Drawings mentioned in paragraph 2 will be furnished to the bidders.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission, to manufacture use or sell any patented invention that may in any way be related thereto.

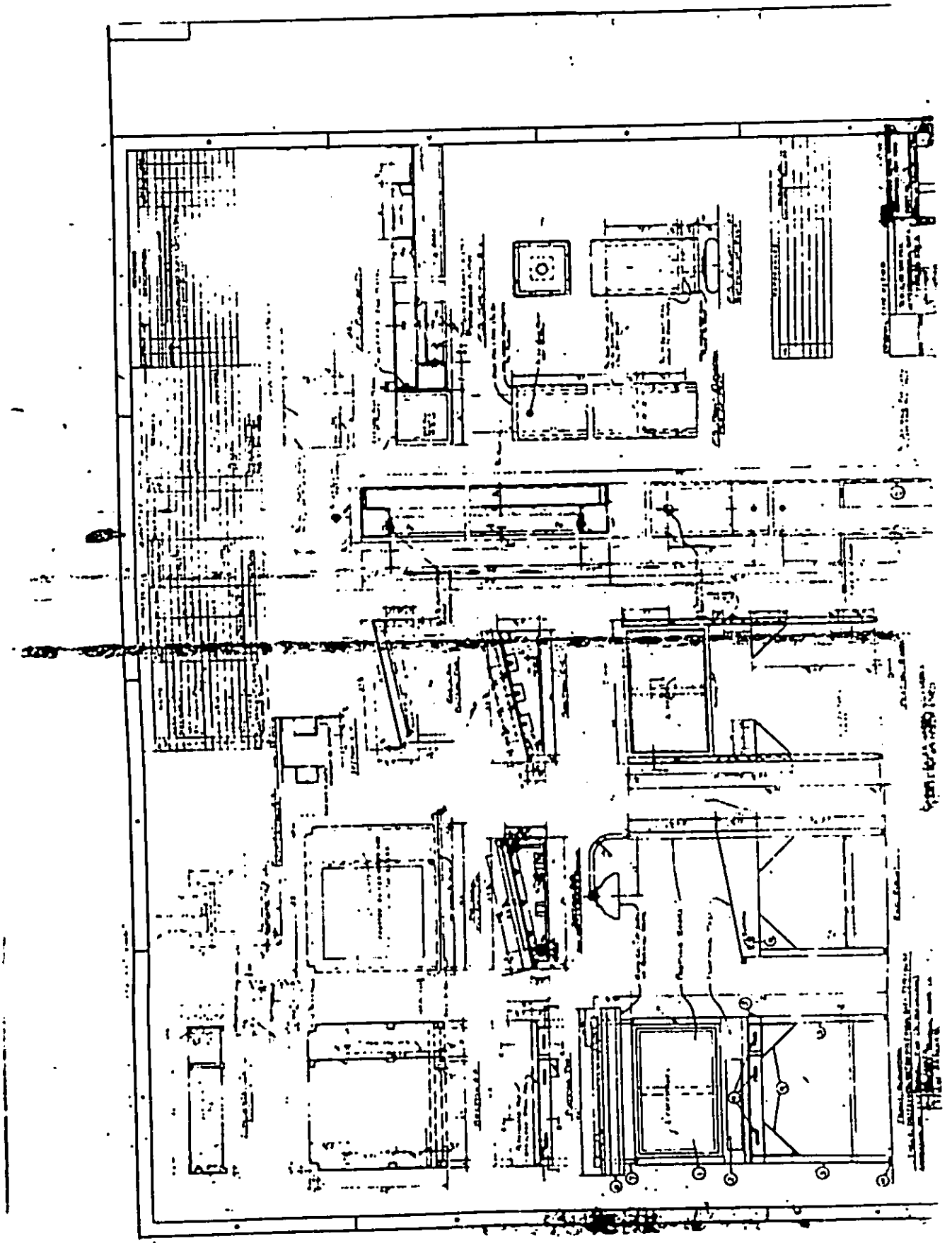
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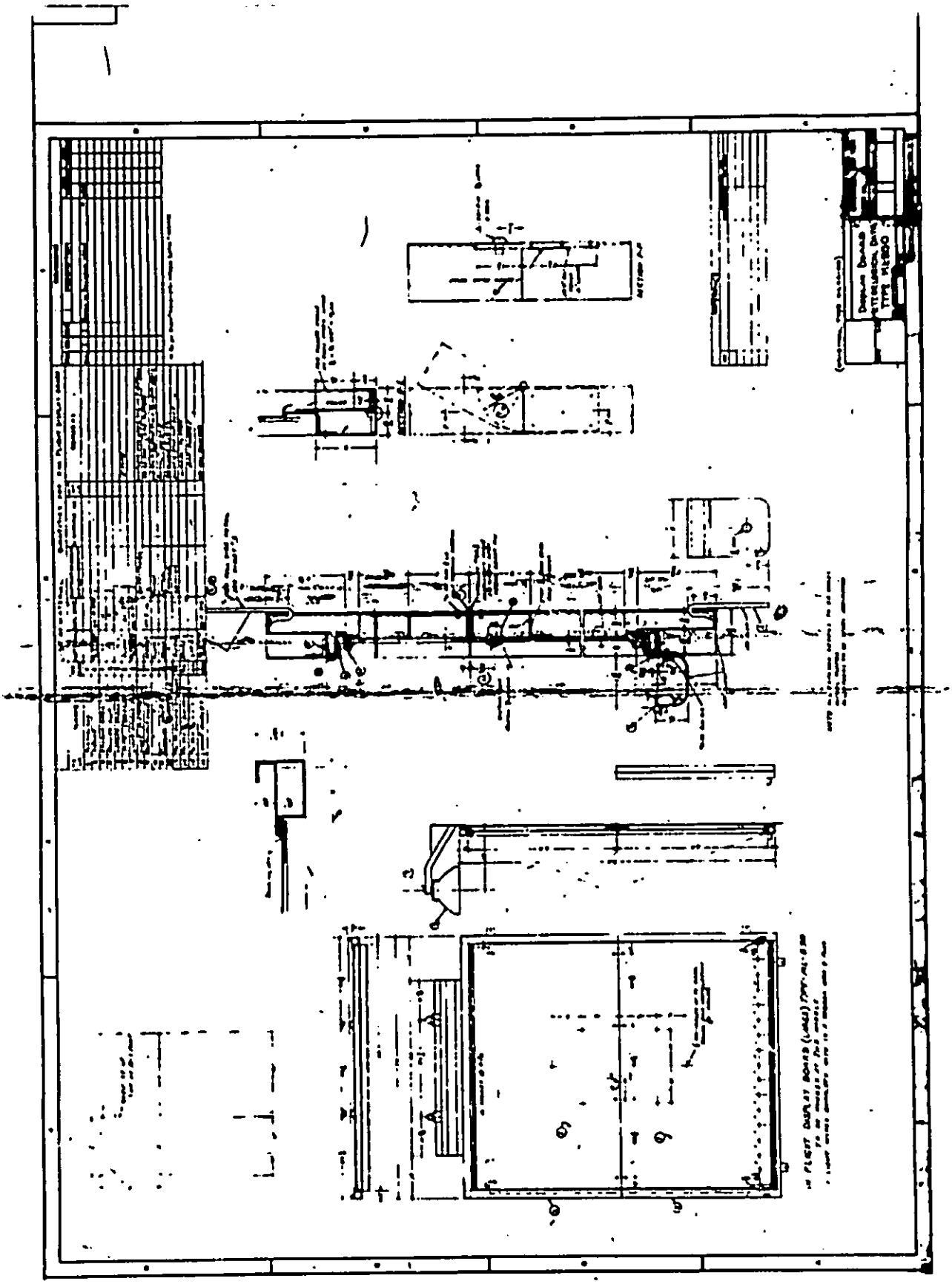
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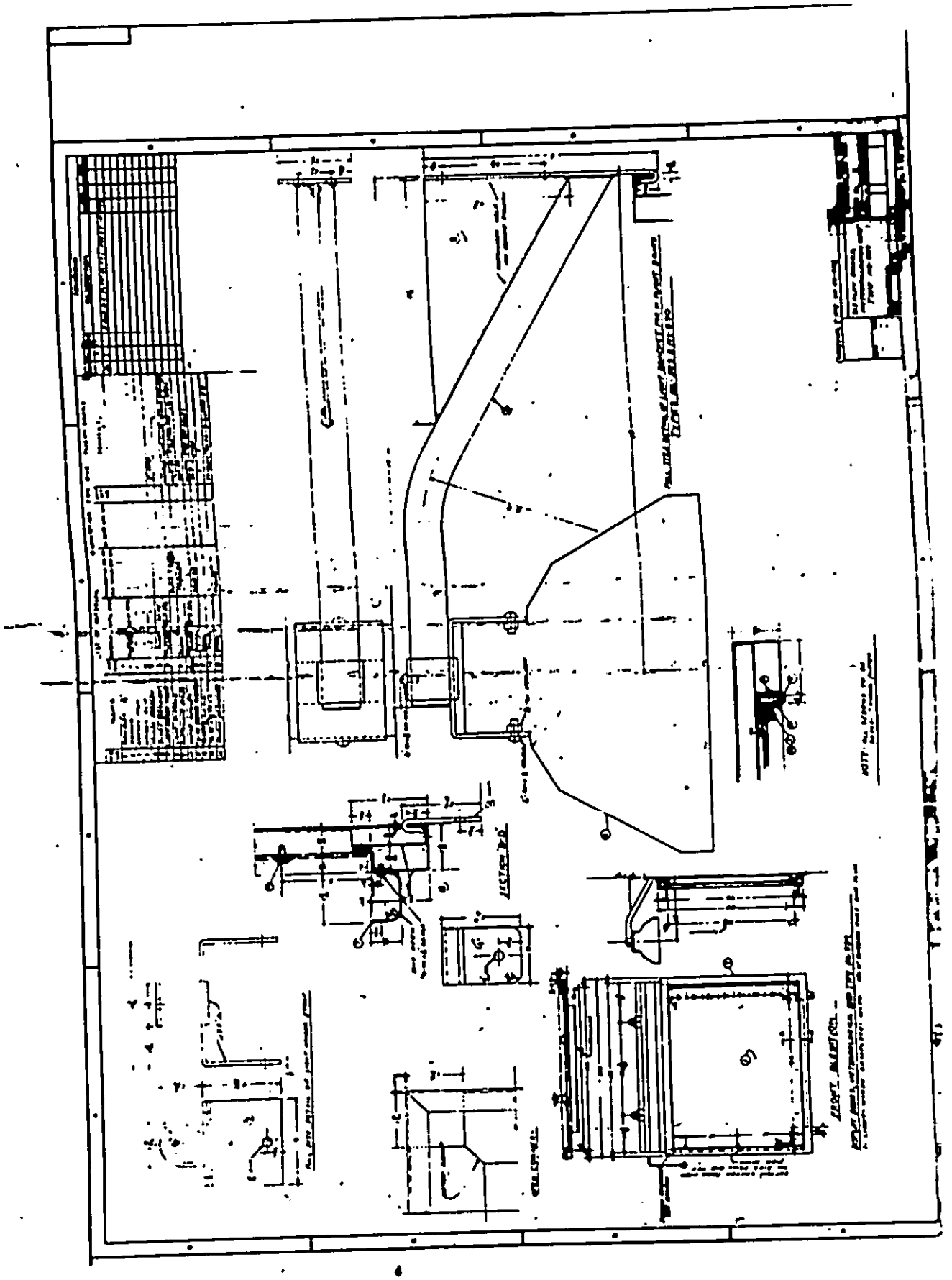
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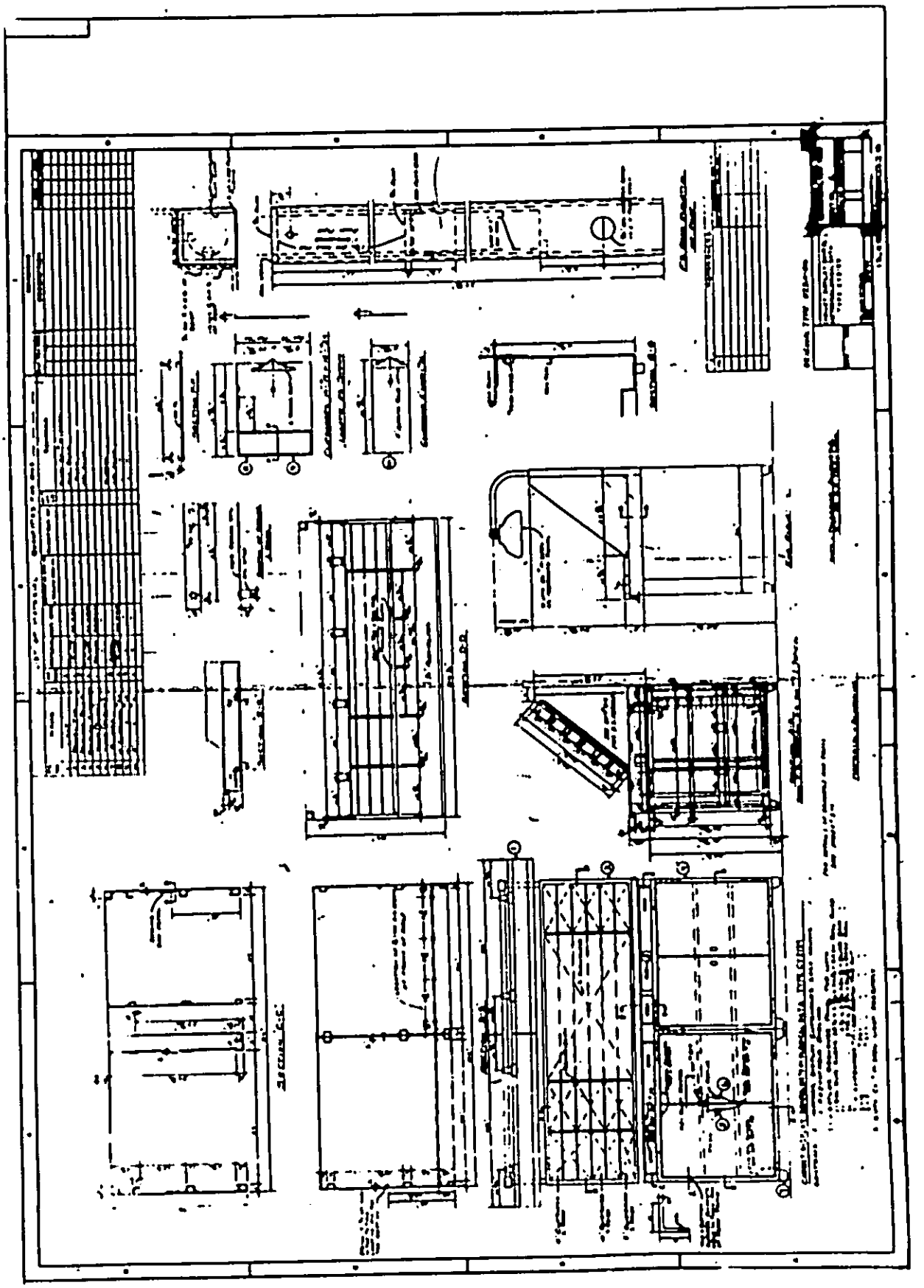
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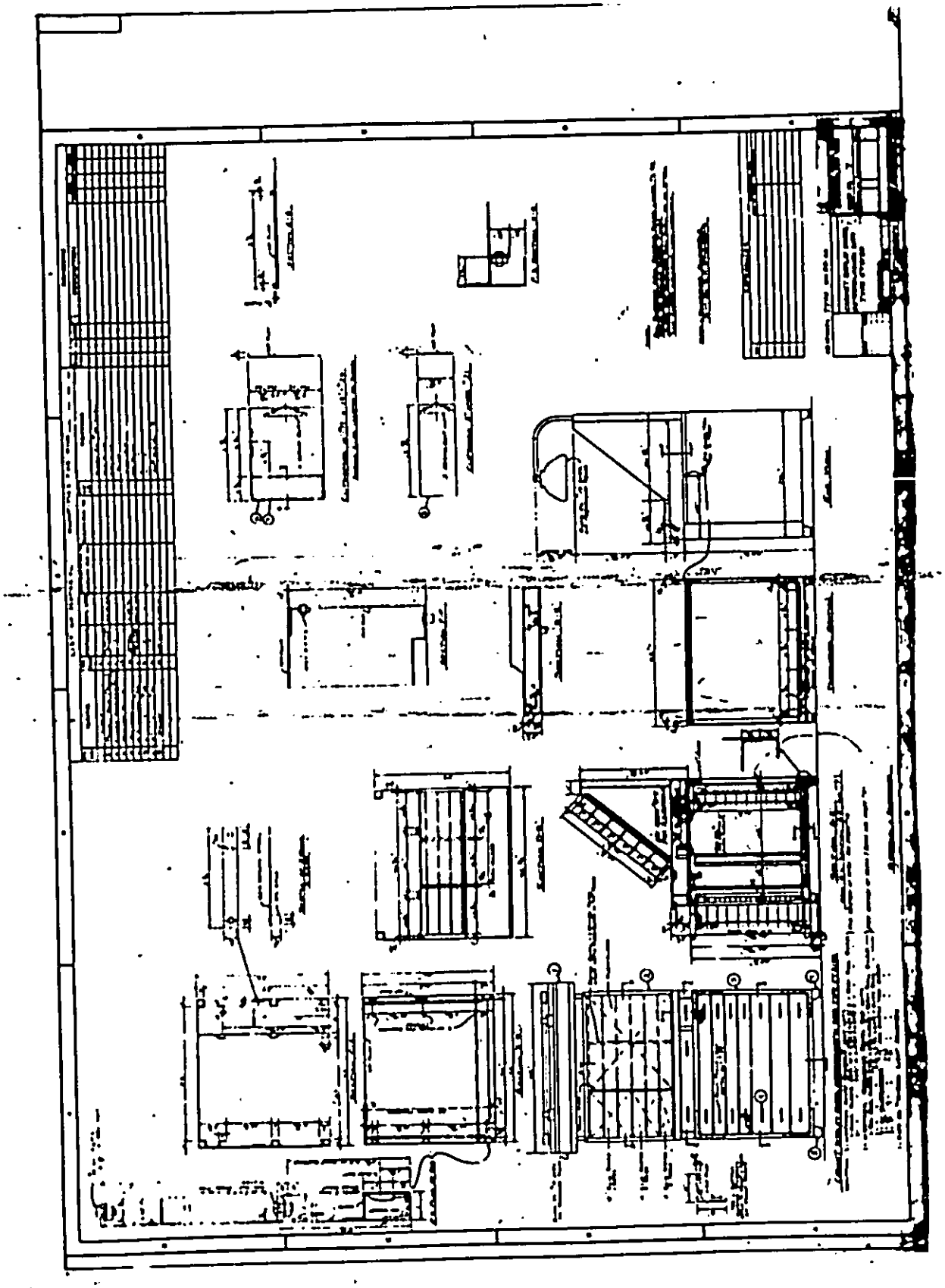
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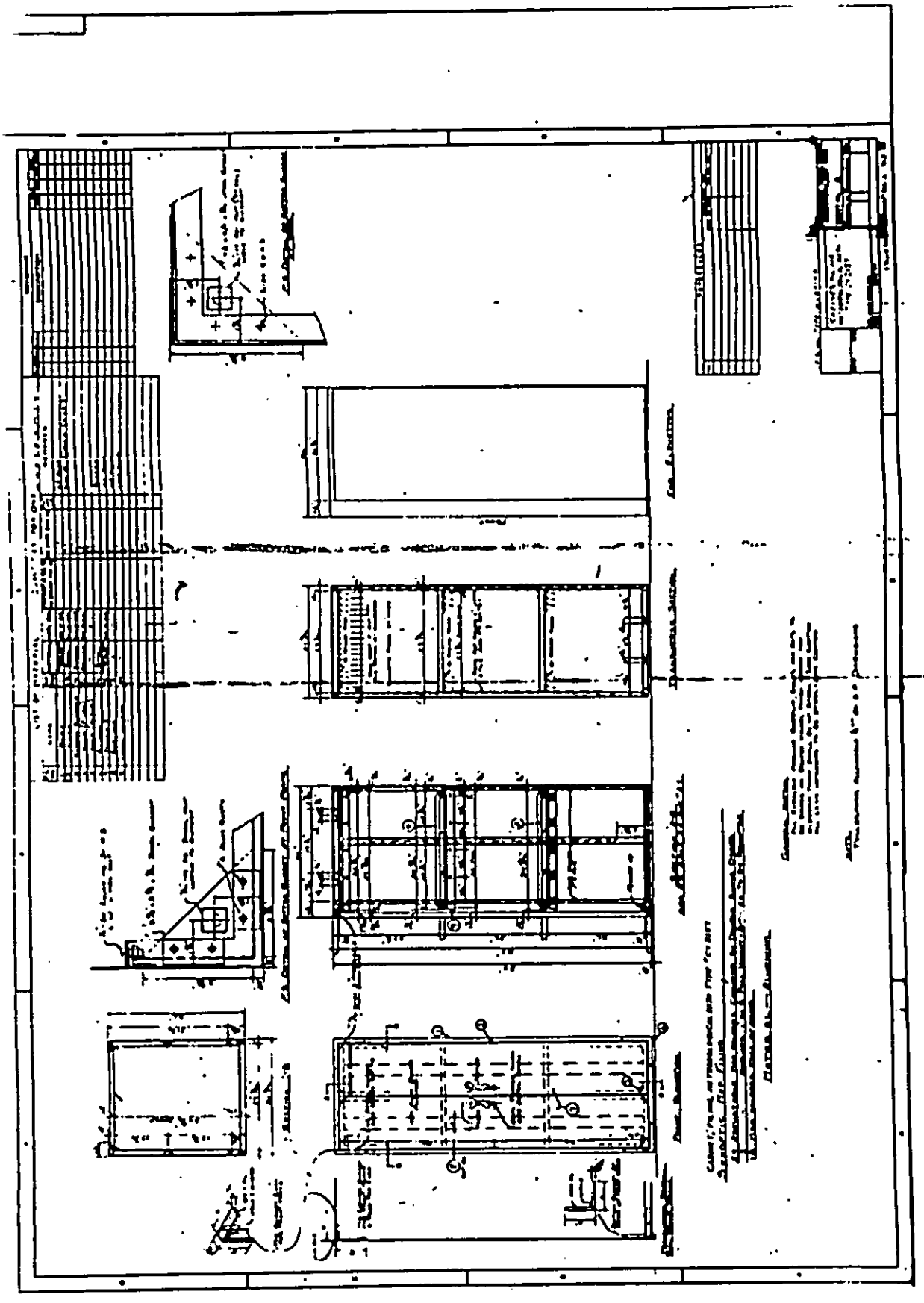


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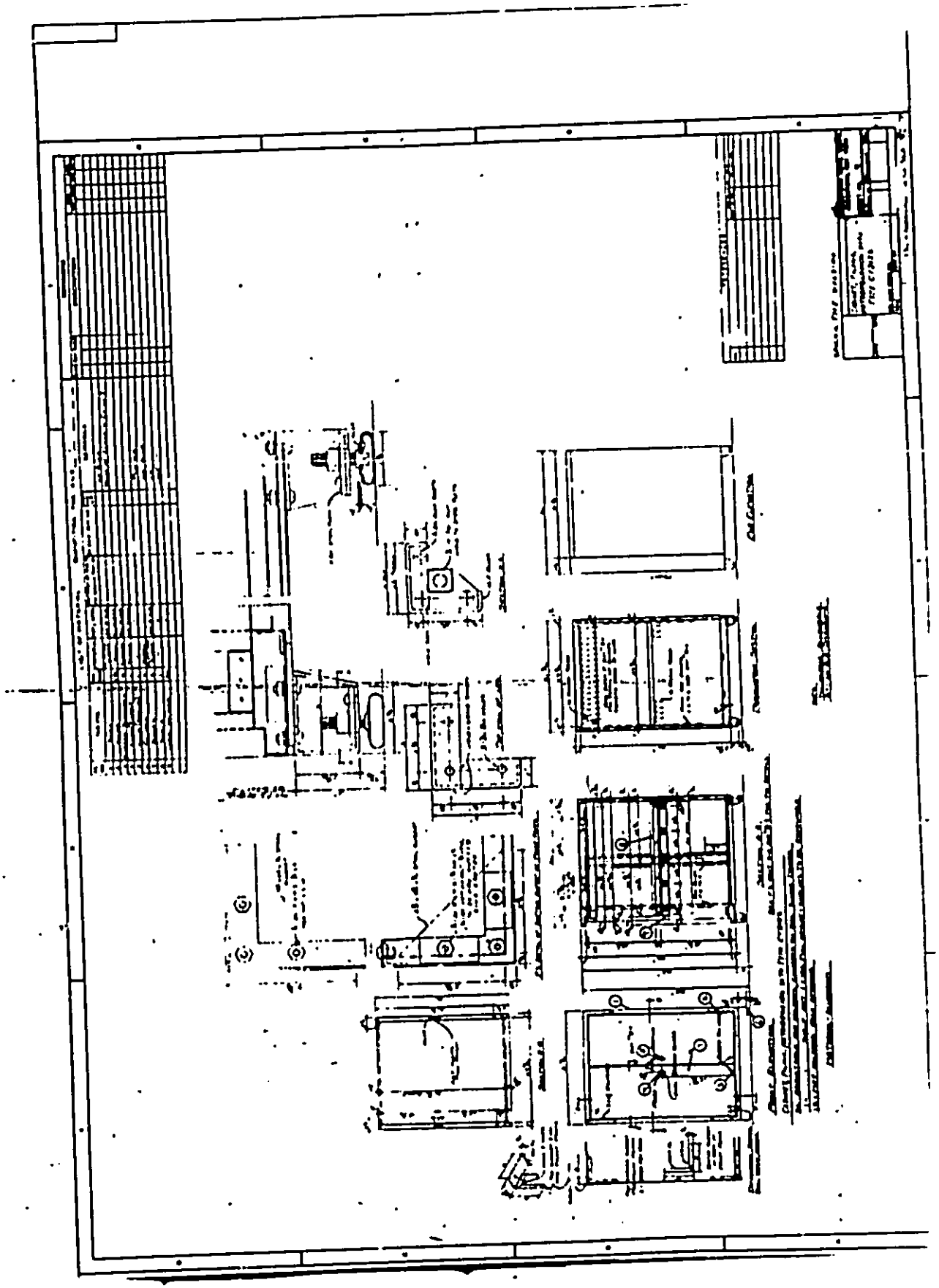


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